

Exercise 8 Collision and Broadcast Domains

Learning Objectives:

- Explain the terms collision domain, and broadcast domain
- Distinguish collision domains and broadcast domains
- Explain devices that connect and separate collision and broadcast domains
- Explain the function of the arp cache
- View the content of the arp cache

Legend:

 Reading that must be completed before the exercise

 Questions/tasks to be worked on before the exercise

 Tasks to be worked on during the exercise

8.1 Preparations for the Exercise

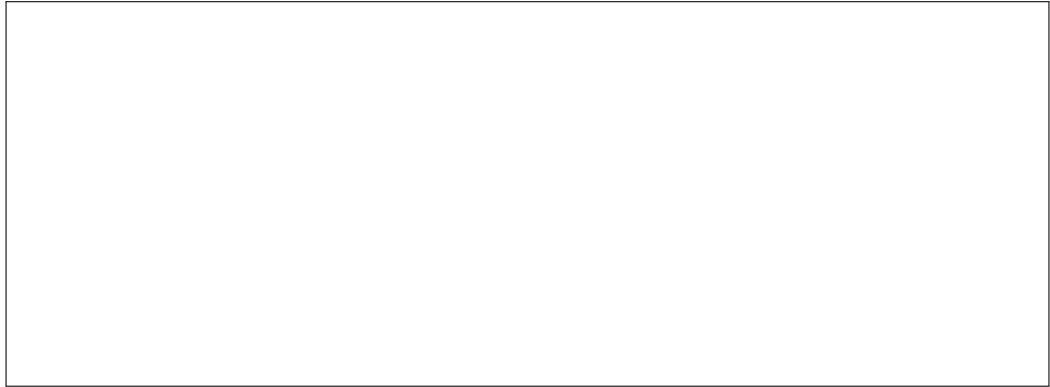
The preparations and questions must be completed before the exercise session. The exception is this first exercise, as you have not yet had the opportunity to read the material. These are also intended as aids to make the tasks that need to be worked on during the exercise session easier for you.

 As preparation, study the following materials:

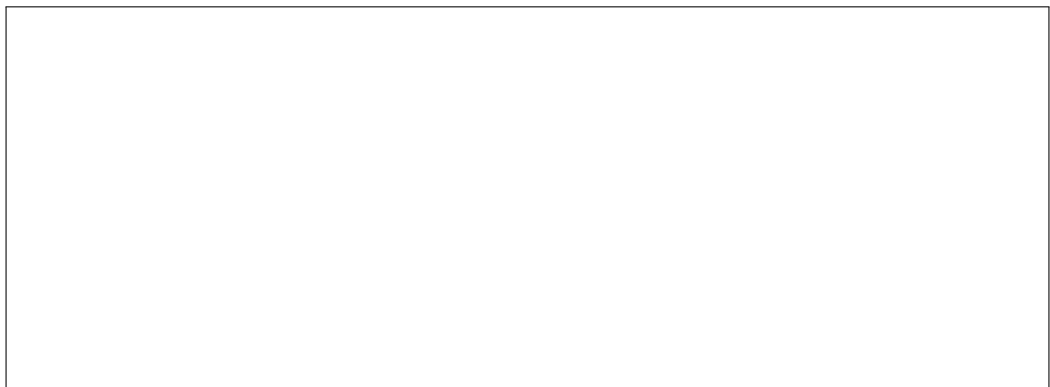
- Chapters 6.9, 7.1 of Christian Baun, *Computer Networks / Computernetze*, Bilingual 2nd Edition, Springer Vieweg, Wiesbaden 2022
- YT-Video: ARP Part 1: ARP Request
- YT-Video: ARP Part 2: ARP Reply
- YT-Video: ARP Part 3: Sending packets to the default gateway
- arp(8) - Linux man page

 **Introductory Questions:**

- a) Name the network devices which divide collision domains.



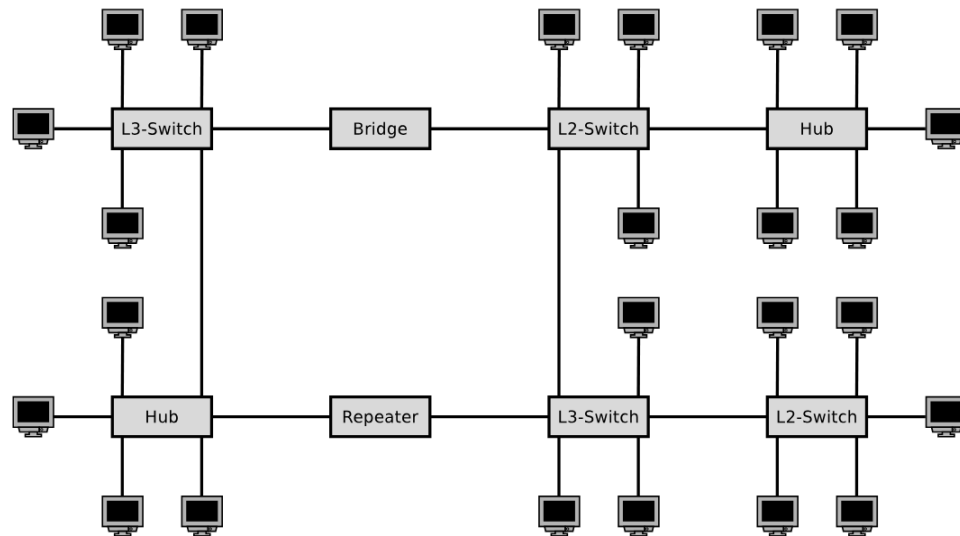
- b) Which network devices **do not** divide collision domains? Name them all.



- c) Name the network devices which divide broadcast domains.



- d) Sketch all collision domains (in dotted lines) and all broadcast domains (in grey lines) of the given network diagram.



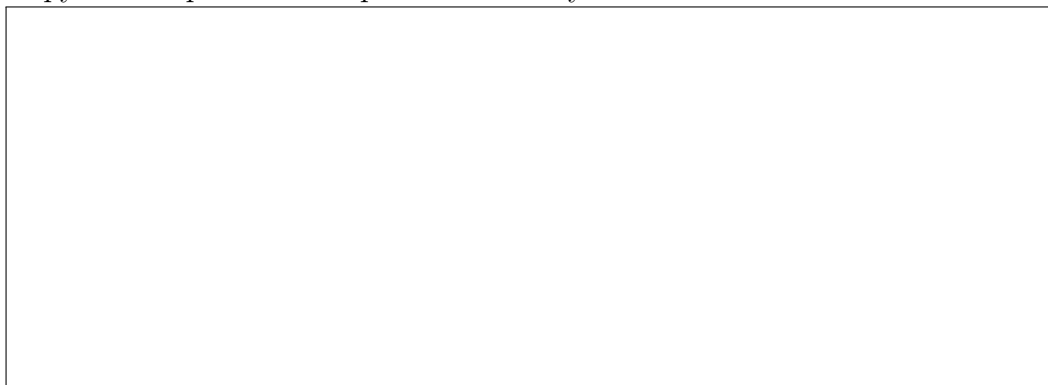
8.2 Address Resolution Protocol (ARP)

🖱️❖ Start Wireshark on your virtual or native Linux system (or Windows system, if you have installed Wireshark there).

- Consult the manual page(8) for arp under <https://linux.die.net/man/8/arp>
- Start capturing the network traffic with Wireshark
- Printout the contents of the arp cache of your local device
- Copy the output of the arp command to the following text section



- e) What do the respective items represent.
- f) Contact the website `www.th-ab.de` with a web browser
- g) Stop capturing the network traffic with Wireshark
- h) What is the output of `arp` now? Do you find an entry for `www.th-ab.de`
- i) Try to explain this behaviour.
- j) Find out the default gateway ip address from the network settings.
- k) Start capturing the network traffic with Wireshark again.
- l) Delete the arp cache entry for the default gateway.
- m) Printout the contents of the arp cache of your local device
- n) Copy the output of the `arp` command to your text



- o) Stop capturing the network traffic with Wireshark again.
- p) What has been changed? Why is the default gateway still in the arp cache?

q) Can you identify the respective request in the wireshark capturing?